RESEARCH METHODOLOGY & SCIENTIFIC WRITING

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ABOUT THE SLIDES

The slides provide an overview, giving links to further reading.

- Many similar guidelines exist and a single right way does not exist.
- Large slide set, not all content is covered in the lectures.

Use as a self-study and reference material.

- The slides are based on recurring challenges I have observed in Master’s theses.

RESEARCH METHODOLOGY

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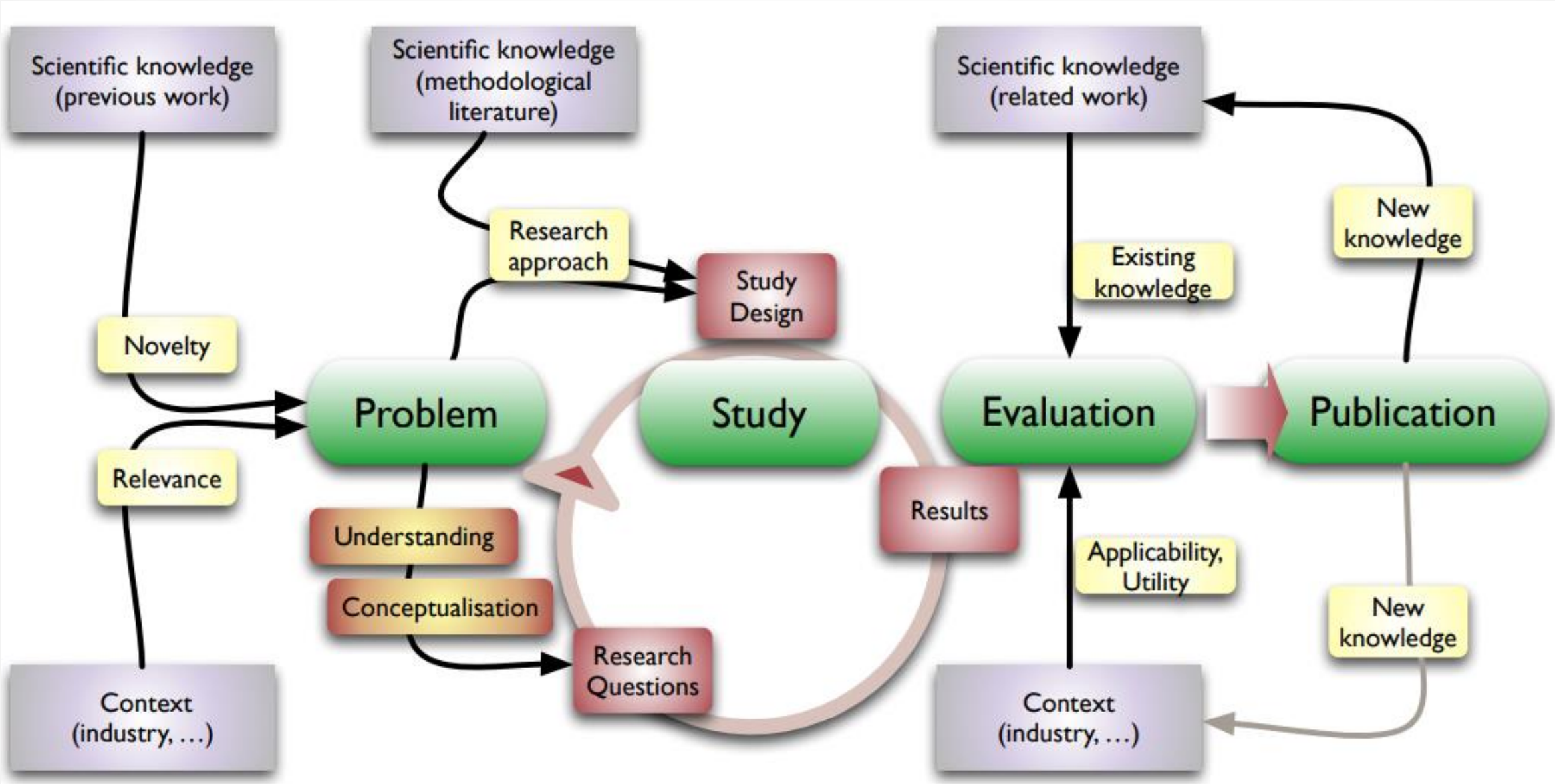
SCIENTIFIC PROCESS



Methods for Software Engineering Research (MSER) Self-Study
https://moodle.helsinki.fi/course/view.php?id=56760

WHY A RESEARCH METHODOLOGY?

Established research methodologies give you a playbook or guidelines how to structure and carry out your research.

- Established methodologies form frameworks that help readers to understand and evaluate your work, even without reading all details.

Required increasingly often in technical sciences.

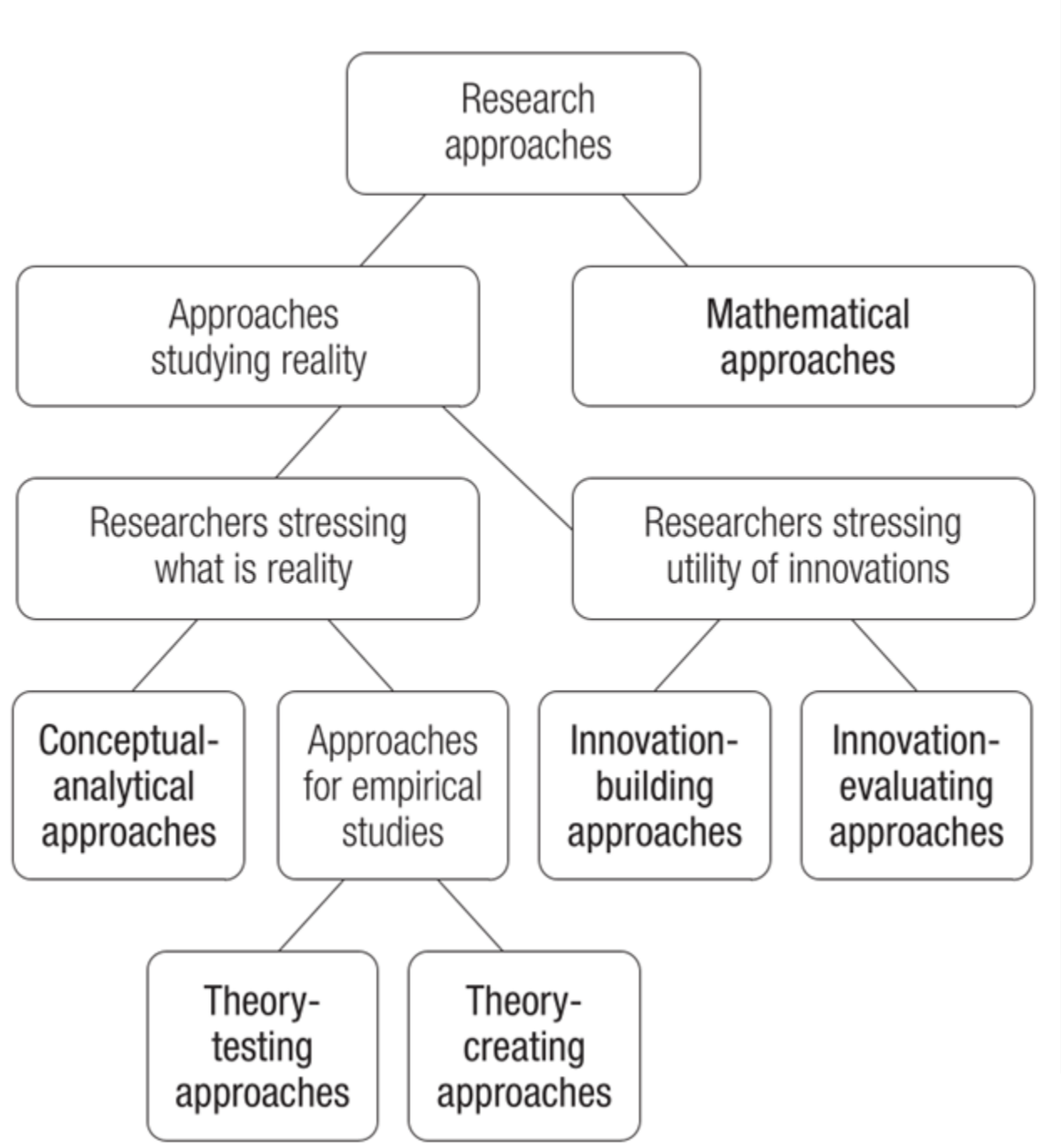
- E.g., “3. Data and research method” part in the MSc thesis assessment matrix, grade 4: “Good and well-argued use of research methods and data, *clearly based on scientific literature or tradition*”.

Scientific writing style and structure are key parts in communicating the research.

- Emphasized also by practitioners when writing and reading technical reports.

TAXONOMY OF RESEARCH APPROACHES

CLASSES NOT FULLY MUTUALLY EXCLUSIVE



Järvinen, Pertti. "On research work in information systems: Guidelines, recommendations and examples." (2021). University of Jyväskylä

CLASSES OF THEORIES/OUTCOMES
(EXCLUDING MATHEMATICAL APPROACHES)

Table 2. A Taxonomy of Theory Types in Information Systems Research

Theory Type	Distinguishing Attributes
I. Analysis	Says what is. The theory does not extend beyond analysis and description. No causal relationships among phenomena are specified and no predictions are made.
II. Explanation	Says what is, how, why, when, and where. The theory provides explanations but does not aim to predict with any precision. There are no testable propositions.
III. Prediction	Says what is and what will be. The theory provides predictions and has testable propositions but does not have well-developed justificatory causal explanations.
IV. Explanation and prediction (EP)	Says what is, how, why, when, where, and what will be. Provides predictions and has both testable propositions and causal explanations.
V. Design and action	Says how to do something. The theory gives explicit prescriptions (e.g., methods, techniques, principles of form and function) for constructing an artifact.

Gregor, Shirley. "The nature of theory in information systems." MIS quarterly (2006): 611-642.

TYPICAL OUTCOMES IN COMPUTER SCIENCE

Conceptual-analytical / Type I:

- A taxonomy, ontology, etc.
  - E.g., DNN cost-efficiency taxonomy DOI:10.1109/ACCESS.2023.3275431

Theory creation or testing / Type II:

- A case study (in a company) how development is done, understanding communication patterns, application of a tool in a company, new problem, or context.
  - E.g., decentralize data platform DOI: 10.1007/978-3-030-91452-3_3
- A systematic literature review (meta-review) what works in which contexts (cf. Type I).

More rarely, predictive Type III-IV

- Typical in natural or social sciences (in situation A, X cause B because C), but CS focuses on purposeful application of innovations (in situation A, applying X leads to B).

Innovation building or evaluation / Type V:

- Method, tool, algorithm etc.
  - E.g., autonomous MLOps pipeline DOI: 10.1109/SEAA60479.2023.00016
- The most typical approach in CS stressing truthlikeness and utility.
- Sometimes comparable to type II (or even III-IV) as an evaluative study.
- Can be about following (or combination):
  - 1. Constructing new or improving existing ones.
  - 2. Evaluating/comparing empirically existing one(s) in specific context (also as theory testing/Type II).
  - 3. Adaptations and applications to new problems or contexts.

RESEARCH APPROACH, METHOD, AND TECHNIQUE AS A HIERARCHY

Research methodology can be understood as a hierarchy of:

- **Approach**: In which class your primary interests and outcomes fall (cf. previous slides).
- **Method**: Which existing research method you applied (e.g., case study, design science, literature review…)
- **Technique**: How were data collection and analysis done?

Not fully mutually exclusive but helps to frame the nature of your study.

- Many-to-many relationship, i.e., approach can be addressed by different methods, and a method can be applied for different approaches etc.

Examples:

- **Descriptive** research applying the **case study method** in a company about development challenges or solutions using **interviews and document analysis** (including Jira, Github) technique.
- **Design and innovate** research of a novel method applying the **design science method** by using **company's data in isolated settings.**
- **Innovation evaluation** research applying **design science method** to adapt, apply and evaluate the utility of the different tool(s) for the specific **problem, data and context,** such as in a company or mobile device**.**

EVALUATION: PECULIAR CHARACTERISTIC OF INNOVATION RESEARCH

Explicit evaluation is needed in constructive (innovation building) research.

- When studying natural phenomena, observation suffice.

Different types of evaluations, e.g.,

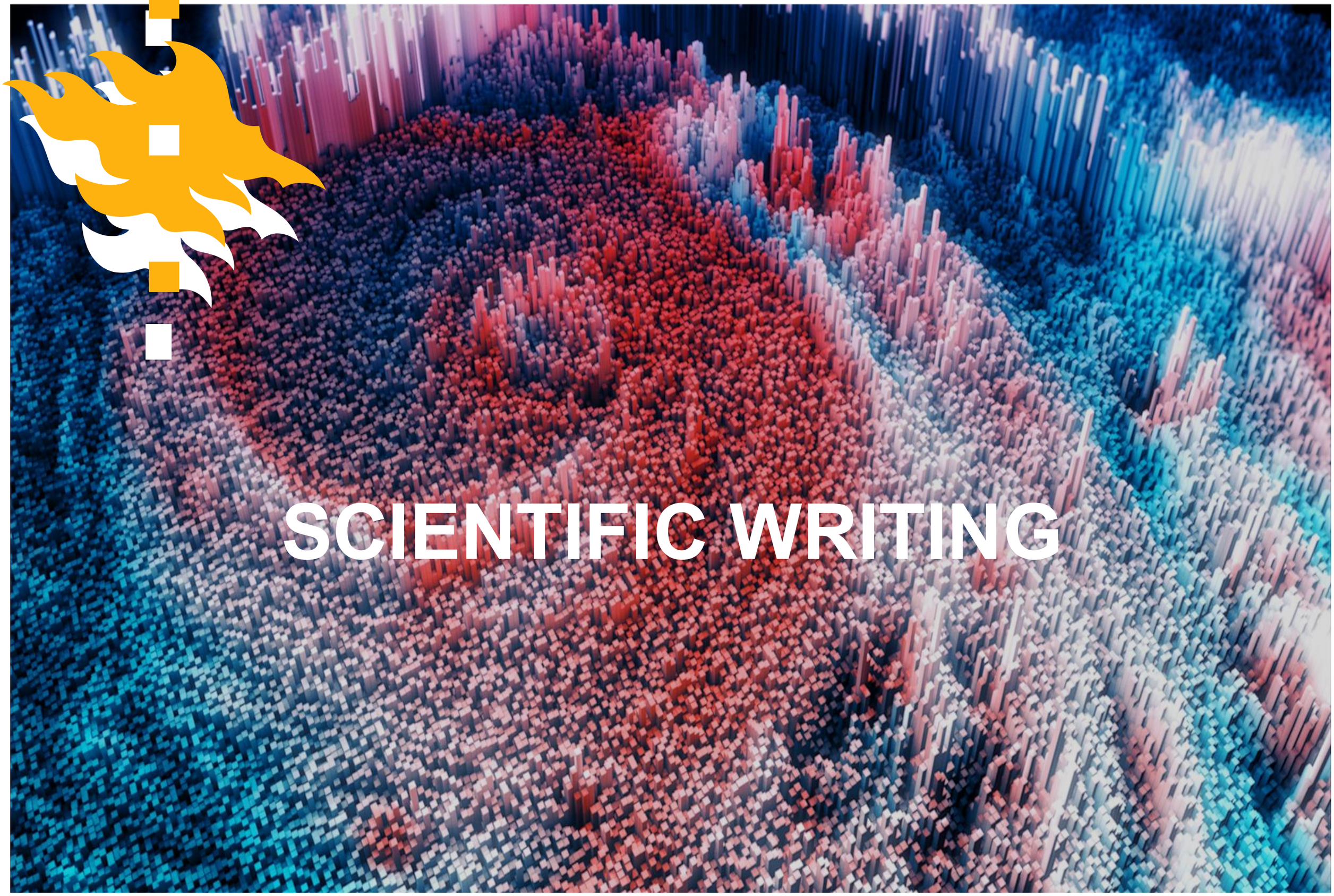
- *Feasibility*: "Hey, look it works/does not crash" -- sufficient only for something really novel.
- *Expert opinions:* from industry, not your own or you friends.
- *Quantitative measures:* using industrial data or open data – representaviness, availability, etc
- *Argumentation, qualitative assessment*: E.g., against SQUARE quality taxonomy.
- *Market test:* "Solution used in my team's operations", also provide measures, opinions, etc.

Define and justify evaluation criteria explicitly.

- *"Lenses of analysis"* through which you assess.
- Reuse existing criteria, such as other paper, standards etc.
- Evaluation can simplify some concerns or settings but be explicit.

Evaluation does not need to proof or show that the solution is perfect but assess it.

(This slide is in progress)



SCIENTIFIC WRITING

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HOW TO READ&WRITE A SCIENTIFIC PAPER

A scientific paper is **not an essay or a story.**

- *"Reading a scientific article isn't the same as reading a detective story. We want to know from the start that the butler did it"* (Ratnoff, 1981 in Gastel&Day, 2016).
- Deliver your message early and then elaborate and give details.
- It should often not be necessary to read all sections!

There are expected structure and conventions in scientific paper, often referred as IMRAD (details follow).

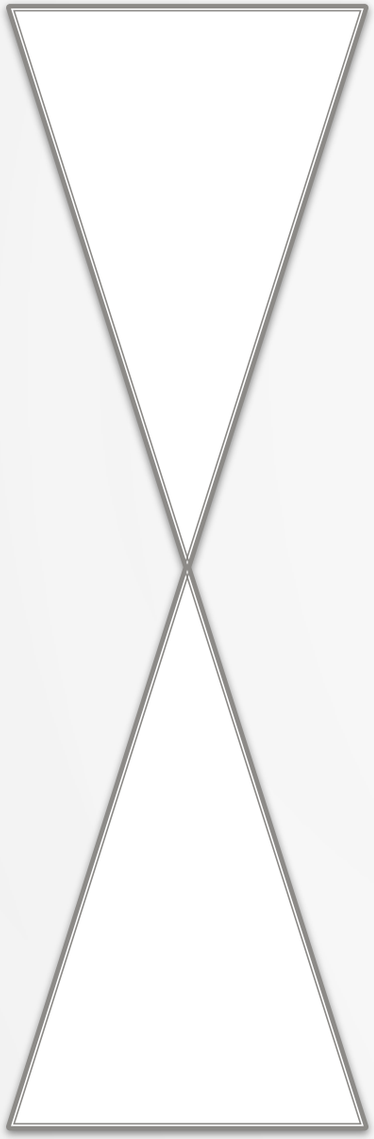
IMRAD allows a reader to understand the key messages by reading only parts of the paper, e.g.,

- Title and abstract should indicate relevance to a reader.
- Introduction and conclusions (discussion) both should state the problem and key findings with different focus.
- Introduction (and background) should introduce the necessary terminology and state-of-the-art what has been reported before motivating and emphasizing what is your novel contribution.

"Borrow" structures from the published papers.

IMRAD STRUCTURE - TIME GLASS MODEL

E.G., HTTPS://EN.WIKIPEDIA.ORG/WIKI/IMRAD
DETAILS OF ALL SECTIONS IN FOLLOWING SLIDES.



Introduction: Start with broad context, key concepts, challenges, and motivation. Narrow down to research problem.

(Sometimes Background or Previous work section)

Method (and your data or materials): Describe what you did to make the study repeatable and assessable.

Results: Focus on your specific findings within the defined scope and context (the narrowest point in the time glass).Often in technical research divided into solution and evaluation.

Discussion: What do the results tell as the answers to RQ(s). Summarize, discuss critically and in broader context, generalize, provide own insights.

(Related work and validity&reliability as a part of discussion or separately)

Conclusions: Summary and contributions to the broad problem, focusing on results, take-away/so-what, and future work.

VARIATIONS TO IMRAD

IMRAD structure is well known and widely applied across different scientific fields.

- E.g., in high-quality software engineering papers, IMRAD was not typical 20+ years ago but has now become almost *de facto*.

It's better to violate the rules, when you know the rules.

However, in constructive research and experimental research comparing techniques or solution, the method section is not always required/own section or very short.

- Often includes a construction and evaluation, and much of method information is naturally placed as an evaluation setup.
- E.g., the research method can be omitted and briefly outlined in introduction and elaborated in the experimentation section.
    - Cf. Results section

ABSTRACT

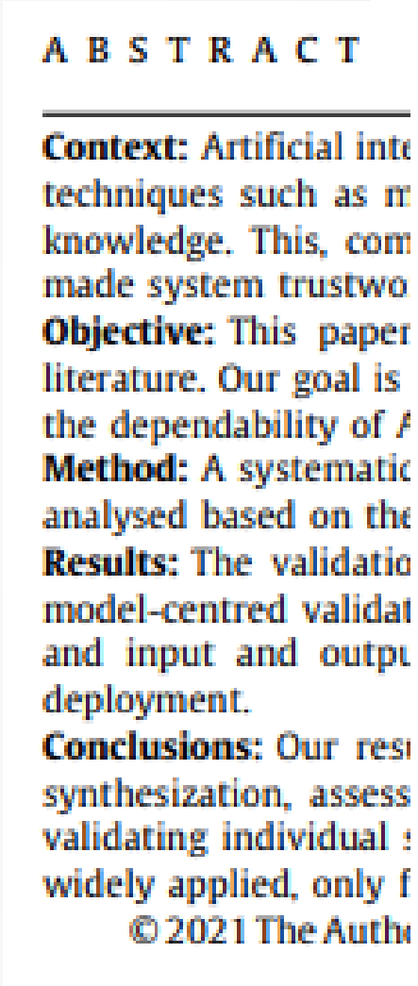
A summary of the paper in 150-500 words (cf. IMRAD).
Typically in past tense.
Some forums require or encourage using a structured abstract

- Even if not exactly followed, gives a good structural guideline.

One structure, variations exist:

- **Context:** Basis, rationale and motivation for conducting the study.
- **Objective:** Goal, research problem, main research question, maybe hypotheses or propositions.
- **Method:** Research method, data or materials the study is based on.
- **Results:** Very brief summary or characterization of the main results.
- **Conclusions:** So what, takeaway message, what to learn from the study, implications of the results. Possibly main limitations, e.g., if crucial for interpreting the results.

E.g. DOI:10.1016/j.jss.2021.111050


ABSTRACT

Context: Artificial inte
techniques such as m
knowledge. This, com
made system trustwo
Objective: This paper
literature. Our goal is
the dependability of A
Method: A systematic
analysed based on the
Results: The validatio
model-centred validat
and input and outpu
deployment.
Conclusions: Our resu
synthesization, assess
validating individual s
widely applied, only f
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IMRAD: INTRODUCTION SECTION

Basic structure of principal components:

1. Background, context and scope for a (peer) reader (*"X is…"*). Shortly.
   - Introduce neutrally the key concepts and problem area (*"Performance refers to ML model accuracy / computing resources use"*).
   - The reader should understand what the paper is about after reading this first paragraph.
2. The existing, comparable solutions and their weaknesses (X is solved by A (Smith, 2000), but…), if any.
3. Gap, problem area, motivation and objective ("However,...")
   - It is crucial to convince reader that paper is relevant.
4. Specific problem ("This paper addresses... The research questions are...").
5. Research method and materials (The research was conducted…").
6. Highlight contributions as principal results and conclusions ("Our findings…").

E.g., one paragraph each

- Or combine, e.g., 1&2, and 4&5, but do not omit.

Often, the structure in the end: "Section 2 outlines background. Research methods…"

PRACTICAL EXAMPLE HOW TO WRITE INTRODUCTION

- First, rather than writing the content, write what you should write there.
  - Add yourself a writing guides as comments (%) in a latex file.
- Mark TBD for the actual text to be filled.
- This example combines 1&2 and 4&5 from the previous slide.
  - Applicable, e.g., to short papers.

```latex
\documentclass{article}
\title{Example intro}

\begin{document}
\maketitle
\section{Introduction}

% CONTEXT AND SOLUTIONS
TBD

% PROBLEM AND MOTIVATION
TBD

% RQS AND METHOD
TBD

% CONTRIBUTIONS
TBD

% STRUCTURE
This paper is structured as follows....

\end{document}
```

IMRAD: BACKGROUND / PREVIOUS WORK SECTION

Optional section that elaborates and refines background or previous work you build on.

Allows shorter introduction, being often necessary in a thesis, but less so in a paper.

Either one or both:

1. **Background concepts** you build your work in more depth allowing delving deeper than in introduction.
   - E.g., difference between validation and verification; or computational vs predictive performance and measures; or AI4SE vs SE4AI.
   - Different definitions for the concept -- with synthesis recommended.
     - E.g., MLOps is defined as X (Author A) and Y (Author B). The similarities are… The differences are... We use MLOps in the meaning of....
2. **Previous solutions** can be described more thoroughly than in introduction.
   - “X is solved by A (Smith, 2000). ….”
   - Summarize how your solution differs either after each solution or in the end of the section. ("Compared to A, our solution....")

Describe neutrally, using present tense.

IMRAD: RESEARCH METHOD (AND MATERIALS) SECTION

Describe the carried research so that it is possible to

- **understand what you did.**
- **assess for validity.**
- **repeat/replicate** in another setting or data (CS research rarely replicates).

Possibly add online resources or appendix.

Describe the research method in past tense, i.e., what you did:

- “We carried out experiment/interview/…”
- Leave the results to the results section.

Describe your materials, such as data, context (lab experiment, an account of company).

- Sometimes materials or data a separate section.

Sometimes bullet point lists helpful although often not recommended elsewhere in scientific writing.

- E.g., when describing computing infrastructure, respondents to interview, etc.

Research methods sometimes omitted in technical papers, but required increasingly often.

IMRAD: RESULTS SECTION

Present the results and findings you have.

- Concept-centered, without verbiage, neutral.
- Use tables and figures.
- In qualitative research, quotations can be helpful and concrete.

Tell only the facts, not your opinions.

- But give descriptions and rationales.

Do not discuss how you obtained your results (it’s the method section), or what else you tried or could have done (it’s the discussion section).

- However, if you properly compared (and evaluated) different approaches, describe both in depth.

Use descriptive section titles.

- Often better to have "Architecture of X" rather than "Results".

RESULTS SECTION(S): CONSTRUCTIVE (INNOVATION) AND EVALUATIVE STUDIES

In technology research, one archetypal approach is to develop and evaluate/demonstrate a solution or multiple solutions.

- Results section often split into 1) description of solution(s) and 2) evaluation.
  - Often further split evaluation into 2a) evaluation setup and data; and 2b) evaluation results.
- Some details of evaluation method and materials (setup and data) can then be in the evaluation section rather than in the research method section.
  - Even omit the research method section and give an overview in the introduction section and details in the evaluation section.

Example sections:

- *Introduction*
- *(Background/previous work)*
- *(Research method)*
- *Principles/architecture of X (and Y)**
- *Evaluation setup and data*
- *Evaluation results*
- *Discussion*
- *Conclusions*

* Different variants, e.g., evaluation setup and data, solution X and its evaluation and then solution Y and its evaluation.

(See the separation of concerns slide later)

IMRAD: DISCUSSION SECTION

Gives answers to the research questions

- Summarize, discuss critically, elaborate implications, generalize, describe alternative solutions you considered...
- Do not downgrade your work or be too modest.
- Remember to emphasize obvious, such as your solution is in production in a company.

Different structures, two typical:

1) Start with overview, then elaborate each RQ, end with comparison to related work, validity&reliability, and future work
   - Applicable, e.g., when you have constructed and evaluated a system or algorithm
2) Cover above parts for each RQ
   - Clear, but requires very clear RQs, such as in a literature review or comparison of techniques.
   - Related work and/or future included in each RQs or separately in the end.
   - Validity&reliability often in the end.

Defined RQs help to make it easy to discuss!

DISCUSSION: RELATED WORK

A subsection in discussion or after discussing each research question.

- Sometimes a separate section.

Related work = Solutions comparable to your results, not background work you build on.

- Often comparing alternative or previous solution mentioned in the introduction of background section.
- Not only how differ but how performs better.

DISCUSSION: VALIDITY AND RELIABILITY

A section of discussion or a separate section/chapter.

- Validity&reliability of distinguish excellent paper from a good one.

Validity ~ correctness, truthlikeness.

- Typically divided into construct, (internal), and external validity.

Reliability ~ repeatability of your study.

- Less often required.

Discuss the validity and reliability threats in your study

- What you did to alleviate them.
- What are implications of weaknesses.

IMRAD: CONCLUSION

Summarise your contributions to the research problem.

Comparable to introduction and abstract. However,

- Summarize your context, problem and method very shortly, 1-2 sentences.
- Focus on principal results, contributions, takeaways.
- Often includes future work, but not limitations of your work.

Avoid repeating the same sentences as in abstract, introduction etc., but use refined wordings.

Referencing to earlier work should have been done in earlier sections.

Often short: 1-2 paragraph or pages in an article or a thesis, respectively.

WRITING TIPS

BASED ON REQUIRING CHALLENGES AND QUESTIONS IN MASTER'S THESES

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GENERAL GUIDELINES

Write to the reader, think from the reader's perspective

Scientific writing favors **formality, clarity, precision, and a neutral tone**.

- Avoid informality, verbosity, contractions (e.g., I'm, don't, can't).

Paper is meant to use **consistent terminology.**

- Be sure that you explain terms (even highlights in italics in the first time) before you use them -- avoid synonyms.
- Avoid synonyms: e.g., if you are using "user", "client" and "customer", clarify and explain the differences.

Scientific results are summarised in **figures and tables**

Check the spelling and grammar with tools

- e.g., https://www.sanakirja.fi/proofing, grammarly.com or LLMs

WRITING IN THE ERA OF LLMS
(IN EARLY 2026)

Producing text has become easier with LLMs

- However, LLMs often produce verbose, descriptive, and abstract text, and complex sentence structures -- unlike expected in scientific writing.
- Typically, you must simplify and concretize the text that LLMs produce.
    - Remove/rephrase adjectives (superlatives) and other similar.
    - Make shorter sentences, exactly to the point. The following sentences then refine and elaborate.

LLMs are good tools for ideation, helping in writer's blocks, and proofreading.

- Do not generate text.

“If you do not time, write a book. If you have a time, write a scientific article”

CONCEPT CENTRIC

Organizing text around key ideas or concepts rather than by authors, or chronologically, to make complex information clearer and easier to understand.

- Concept X (author A, author B,).. Concept Y (author A, author C,).

- Rarely chronological, but sometimes applicable to key concepts: X was developed 2010 (author A), Y improved X 2015 (Author B)…
- *Not* what authors did (author centric): Author A concept X, concept Y,.. Author B concept X, concept W.

If you developed a system X, present the final design and cover in discussion what were alternatives.

- Do not make a diary-like description of the development.

If you tested different approaches, report them concept centrically to show what are approaches and test results

- Even if you did and tested A first, only after that B, you do not need to cover when they were tested if using the same (or similar) setup.

Recommended reading: Webster, Jane, and Richard T. Watson. "Analyzing the past to prepare for the future: Writing a literature review." MIS quarterly (2002): xiii-xxiii.

SEPARATION OF CONCERNS FOR WRITING

Separation of concern is a well-known design principle applicable to scientific writing that divides into distinct sections, each addressing a specific concern (cf. IMRAD).

- Sections (or subsections) are self-contained blocks, understandable in isolation when the key concepts are understood.

Improves clarity, maintainability, and allows parts of text treated, written, and organized independently, without affecting other parts.

- E.g., Often in MSc thesis reordering the description of context, such as company, to be placed before or after the method section (or as a subsection).
- In evaluative studies, allows reordering setup, data, evaluation results parts easily.
- Allows phased writing, i.e., write one block in one day, and another block in another day, and writing some parts even before full study is done.

PARAGRAPHS IN SCIENTIFIC WRITING

- Each paragraph should focus on one (or few) main topic, idea or message.
- Begin a paragraph with a **topic sentence** that states the main point, often followed by references.
- Follow the topic sentence with description or explanation sentences that support the main point.
- Use logical flow: move from general statements to specific details.
- Sometimes end the paragraph with a concluding or linking sentence, when appropriate.

- Paragraphs should be rather too short than too long.
- Paragraphs should be logically ordered, though transitions from one topic to another in scientific writing can be less smooth than in general literature.
  - E.g., when explaining central topics.

HOW TO REFERENCE (IN CS RESEARCH)?

Lorem ipsum (Smith 2023)... Then elaborate in following sentences and a reader understands that the explanation is based on the references.

- The first topic sentence should summarize and following sentences (or even paragraphs) elaborate and do not need repeating reference.
- Allows also multiple topic sentences and references in one paragraph.

X method/tool/algorithm (Smith 2023) lorem ipsum… emphasizes the method (not the author or paper) and the key reference.

- Good style especially when you discuss about specific method/tool/algorithm.
- Reference can be homepage of a tool where all information is available, not needing to reference all sub-pages.

Lorem ipsum. Lorem ipsum. (Smith 2023) in the end of paragraph, after period, refers to the beginning of the paragraph (or previous reference but rarely used in CS).

- Often not recommended but possible.
- Rather use the other styles by refining your text.

Smith (2025) Lorem ipsum. Emphasizes the author that is often not necessary.

- See next slide for details

ABOUT USING REFERENCES

- Smith et al. (2012) write in their publication that pink is a challenging colour.
  - Pink is a challenging colour (Smith et al. 2012).
- Using an author's name in text typically indicates a reason for doing so, e.g., to highlight it is Smith who said so or that is what Smith wrote
  - … because you tend to disagree, there is something controversial, an opinion you want to highlight, seminal contribution
  - The reason easily misunderstood, making using author name often unnecessary.
- More important to reveal the context or details relevant to the message
  - The participants of a boxing event considered pink a useless colour (Smith et al. 2012).
  - The developers in a Finnish game company categorised pink as a positively challenging UI colour (Smith et al. 2012).

RESEARCH QUESTIONS (MY OPINION)?

In quantitative or natural science studies, carefully worded research questions are quintessential.

Especially constructive (innovation) research often has a set of objectives.

- However, formulating research questions help to scope the research and in writing the discussion section.

Research problem and/or questions?

- Research problem (broad) can be divided into RQs (specific) you provide answers.
- Research problem can be formulated as a research objective.
- Discussion answers to the RQs, conclusions to the research problem.

WE OR I? OR PASSIVE?

In English, typical scientific convention is to use “we”.

- “I” might sound arrogant emphasizing you.

When referring to you as authors:

- “The first author carried out interviews alone“ (if you want to be explicit).
- We conducted (even if only one author).

Passive is also often used.

- Possible, but not recommended as a primary style.
- Passive often convenient to avoid emphasizing a paper or author.
- Sentence structures in passive become often more complex making understanding more difficult.

SECTIONS, TABLES, FIGURES…

Papers and chapters have sections -- books (and theses) have chapters.

Section 3, Table 4 etc. are “names”, i.e., capitalized when numbered, but without numbers not:

- “In this section”,.... “...shown in Section 2”

Table captions above tables, figure captions below figures.

- Always remember to reference in the text to the tables and figures
- “Table 2 summarizes….”

See a style from a high-quality journal paper and follow the same style.

- Consistency is more important than selected style.
- My favorite environment in Latex is tabularx that, e.g., automatically wraps text to multiple lines and allows manual tuning of columns.

SERIAL COMMA

Use of serial comma typically adds clarity in the lists of things before “and”/”or”.

- Orange, apple, and banana.

- https://en.wikipedia.org/wiki/Serial_comma

ITALICS TO DEFINE TERMS: OPTIONAL BUT ADDS CLARITY

When introducing a specific *concept* and term for the paper, use italics to “reserve” and highlight that the concept is central when it appears the first time.

- Define or describe a specific meaning for the term near its first appearance.
  - E.g. “Today, software is developed in *agile* fashion. Agile means…”
- Italics only in the first appearance (excluding abstract).
- Only for key concepts.

To test the clarity of you text:

- Make sure the terms in italics are described or defined just after they are mentioned the first time.
- In one paragraph, there should not be too many words in italic (often just one).
- In the introduction section, do not define too many terms (e.g., max 7).
  - You can have a separate background section to further describe the concepts and define additional key concepts.
- Only emphasize key terms, if you end up emphasizing dozens terms, you probably have not conceptualized and focus appropriately.

THIS, THESE, IT, THEY…. I.E. PRONOUNS

Scientific writing aims at precision and using pronouns is often not good style.

- A pronoun in text often indicates unclarity or ambiguity.
- E.g., “The team switched to a new sprint planning tool, but it confused the developers because they thought this would make things harder.”
  - "it” → does this refer to the tool or the switch?
  - “they” → does this refer to the team or the developers (if those are separate groups)?
  - “this” → does it mean the new tool or the process change?
- However, pronouns can be used to emphasize, such as “this tool/this section”.

Use also specific verbs.

- E.g., “is” can often be replace by more precise verb.

Pronouns and general terms can often be replaced by more precise terms.

- “There is” and similar structures can often be replaced more precise wording -- LLMs can help.

A large number of pronouns, is an indicator for bad style (cf. bad smells).

TENSE IN SCIENTIFIC WRITING

Existing knowledge is referred in present tense.

- “Earth is flat” vs “Earth was flat”.
- When emphasis on research activity: “Aristotle showed that earth is not flat”.

Your results is not yet existing knowledge, use past tense

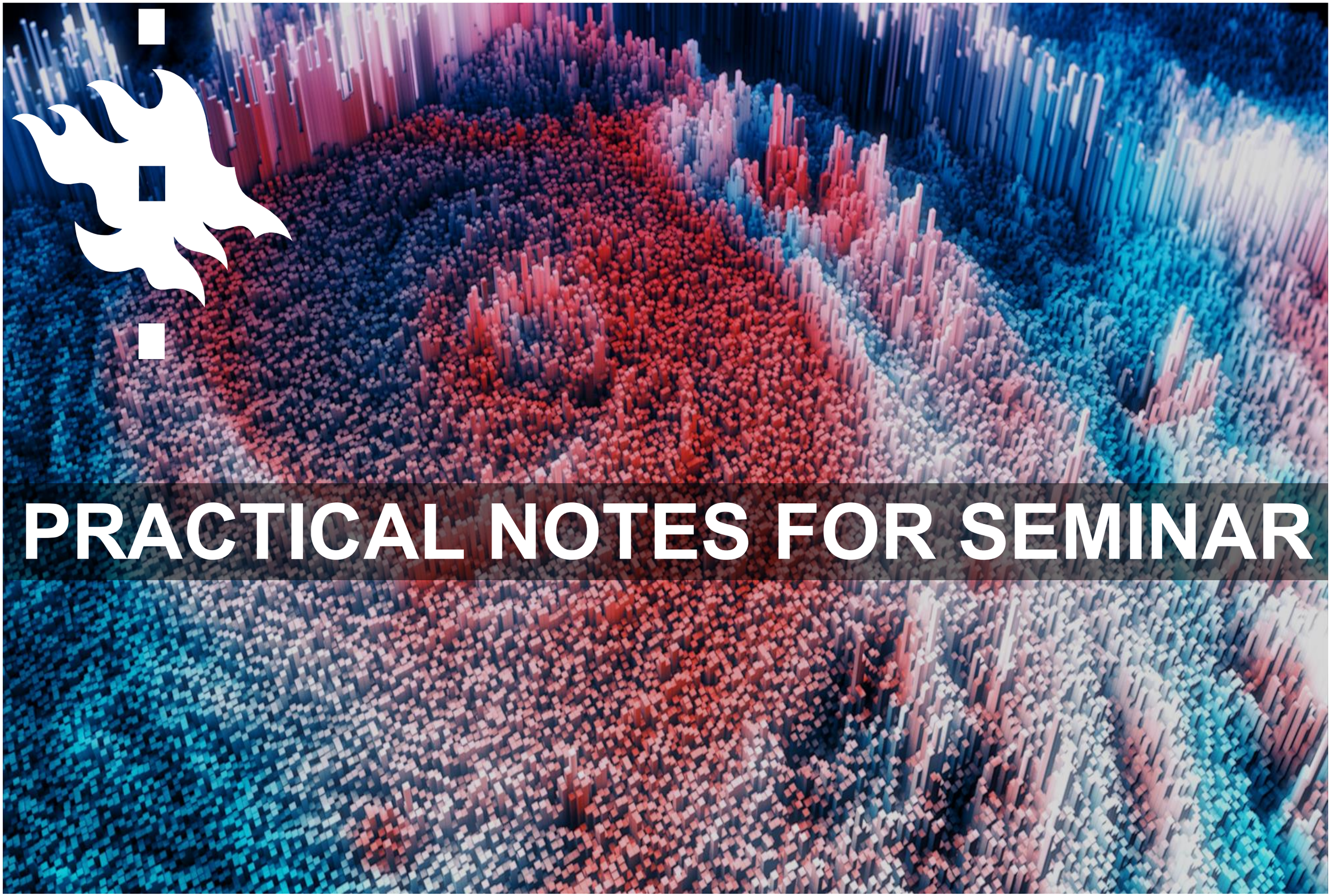
- “Accuracy was 99%”, “we carried out interview”, “we tested”
- However, in CS, present tense is often applied: “Accuracy is 99%”

If your cross-reference within the paper, use present: “Table 3 shows”; “discussion summarizes”, not the future or past tense.

In your paper, typically

- Abstract and method in past tense.
- Introduction and discussion in present, except when referring what you did (“our test showed”).
- Results either past or present tense.
- Very rarely the future tense anywhere.

Recommend reading: Tense chapter in Barbara Gastel and Robert A. Day: How to Write and Publish a Scientific Paper.



PRACTICAL NOTES FOR SEMINAR

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OBJECTIVES OF SEMINAR PAPER

Adheres to scientific paper practices

- Right structure and writing style
- 5-6 pages, additional pages allowed as appendix, e.g., tables of experiment results.
  - It’s more difficult to write focused 6 pages than longer, especially with verbose LLMs.
- Use figures and tables!

Follow the IEEE 2-column template.

- However, it is more important to have good results, to be consistent, and to have a polished look.

With minor modification, the best papers could be submitted to a scientific publication.

SOME PRACTICAL NOTES

Pace yourself -- define time budget

Note that you can “modularize” your work but do not leave it halfway.

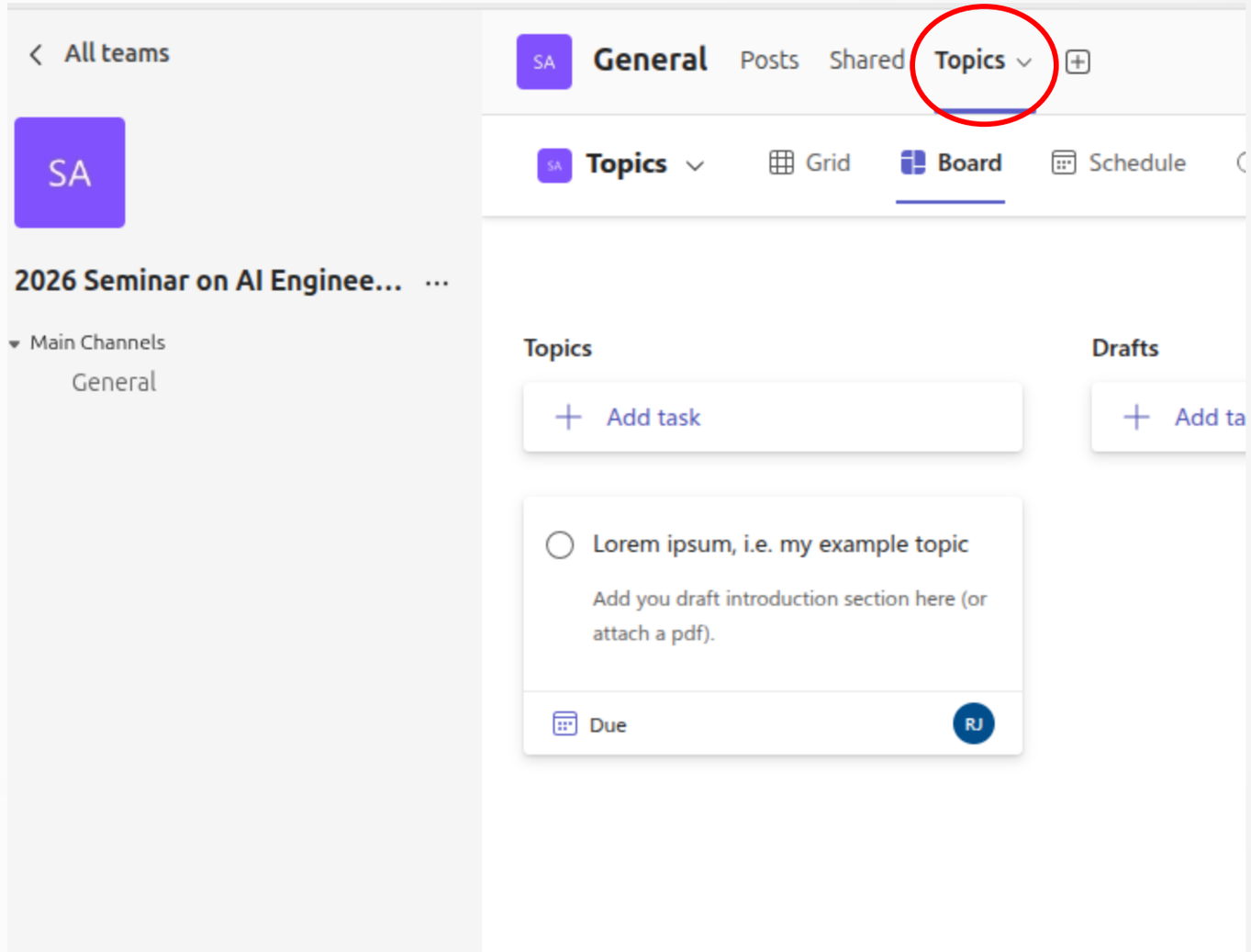
- Cf. IMRAD structure.
- E.g. carry out all experiments/searches at once, write method section
- …but scientific writing is inherently iterative.

“If you do nothing for over 2 weeks, it’s like restarting”.

NEXT SESSION: JANUARY 28

Add your topic card to planner in Teams

- Title
- Introduction draft following recommended structure in the lecture slides.
  - Even if you do not yet have results, elaborate the nature of results.
- Assign the card to you.
- Deadline: The day before lecture (Jan 27)



During the next session, we will review the topics in two groups and rooms.

- Exactum and Chemicum.
- Assignment to rooms in Planner.

HELP, Q&A ETC.

- Weekly zoom sessions for Q&A at the lecture times, when no on-site lecture
  - Starting at 14. Not available entire 2 hours.
  - We'll get notification from zoom be email once you join in.
  - https://helsinki.zoom.us/j/62076155779?pwd=aX0L7kYraM5DiUAPUm5YgNgCYjnp2r.1

FURTHER LEARNING

HELSINGIN YLIOPISTO
HELSINGFORS UNIVERSITET
UNIVERSITY OF HELSINKI

GENERAL

Methods for Software Engineering Research (MSER) Self-Study in Moodle.

- https://moodle.helsinki.fi/course/view.php?id=56760
- Additional references in the end of the slides.
- Not required in this seminar, but helpful for a thesis.
- Includes similar introduction and descriptions of specific methods.

Focus on empirical research.

- Familiarize yourself with the method relevant to you, but understanding all methods is often not relevant.
- Experimentation, such as comparing two approaches, tools, algorithms or applying and testing an approach is not explicitly covered but, e.g., as a part of Design Science.

GENERAL

- The guidelines of ESE group.
  - Based on experiences in software engineering research.
  - https://www.cs.helsinki.fi/group/ese/ScientificWritingGuide.pdf

- At least skimm it through to get familiar with it…

  … and while writing, (re)check specific information.

- Plus: Other similar guidelines.

WRITING

Barbara Gastel and Robert A. Day: How to Write and Publish a Scientific Paper.

- Several editions, the first editions written alone by Day.
- Overall, good and easy to read book, but also for looking practical advices about specific issues, such as tense.
- However, some parts not relevant for CS or outdated.

Tips for Writing Technical Papers: https://cs.stanford.edu/people/widom/paper-writing.html

- Very CS/Data science specific guidelines